Graduation Project Guideline: AI for Healthcare

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Your team of five will design, build and pitch one AI system that solves a real healthcare problem. Each member owns one workflow, the team connects the five into a single working system, and you present it as a business solution.

Getting started

The project turns what you learned in the eight course modules into one working product. You finish with three things:

- **One integrated system** built in n8n, made of five connected workflows.
- **Short documentation** that explains the business problem, the design and the test results.
- **A business pitch** with a live demo.

Ground rules

- **Team of five, one workflow each.** Every workflow has exactly one owner.
- **One problem, one system.** The five workflows must connect. Five separate mini-projects do not pass.
- **No real patient data.** Use synthetic or fully de-identified data everywhere: in n8n, in prompts, in slides.
- **AI drafts, a clinician decides.** Any output that could affect care or reach a patient passes a human approval step.
- **Build in n8n.** You may add other tools from the course if they connect to your n8n workflows.
- **Pass each checkpoint before you move on.** Every phase ends with a short review with your instructor.

Team roles

Each member is a workflow owner and a presenter. Three members also take one shared role.

Role	Held by	Responsible for
Workflow Owner	All five members	Designing, building, testing and documenting their own workflow
Team Lead	One member	Running the team meetings, tracking progress, contact with the instructor
Integration Owner	One member	The integration map and the connected system; approves any change to an input or output

Role	Held by	Responsible for
Evaluation Owner	One member	The LLM-as-judge, LangSmith tracing, the monitoring dashboard and the alerts
Presenter	All five members	Presenting their own workflow in the pitch; the team prepares the slides, the demo script and the rehearsal together

How you work

- Meet online regularly. Each member reports: done, next, blocked.
- Keep everything in one shared folder and one shared n8n workspace.
- Write down every decision that affects another member's workflow.

Phases and deliverables

Phase	What you hand in at the checkpoint
Phase 1: Business Discovery	Business Brief (one page)
Phase 2: Solution Design	Solution Design: features list, five Workflow Spec Cards, integration map, safety plan, evaluation and monitoring plan
Phase 3: Build	Five working workflows and five one-page workflow documents
Phase 4: Integration and Testing	Working end-to-end system with monitoring dashboard and alerts, Test Report and backup video
Phase 5: Business Pitch	Presentation slides and live demo

Phase 1: Business Discovery

By the end of this phase, your team has agreed, in writing, on one problem worth solving and how you will know it is solved. Nobody builds anything yet.

Course link: Module 1, Session 2 (Clinical Workflows and Problem Framing).

What to do

1. **Business problem.** Choose one problem from your own practice. Say where it sits in the patient journey (admission and triage, diagnosis, treatment, discharge and follow-up), who is affected, and what it costs today in time, money, errors or patient experience. Give at least one number, even an estimate.

2. **Problem statement.** Write one paragraph using Template A1.
3. **Business objectives.** Write two or three objectives. Each one is measurable, for example "cut the time to prepare discharge instructions from 20 minutes to 5".
4. **Target audience.** Name three groups: the users who operate the system, the people who benefit from it, and the buyer who decides and pays. Add a short persona for the main user: role, setting, what they do today, what frustrates them.
5. **Success metrics (KPIs).** Choose three to five KPIs across the three lenses from the course: outcomes, cost, and workforce well-being. For each, give the baseline today, the target, and how you will measure it in testing. Use Template A4.
6. **Scope.** List what is in and what is out. State the AI's role as assistive, and list what the system must never do.

A good project idea

- ☐ Is large enough to split into five connected workflows
- ☐ Has one clear main user
- ☐ Can be built and tested with synthetic data
- ☐ Keeps a clinician in control of every clinical decision
- ☐ Can be shown end to end in a short demo

Deliverable and checkpoint 1

A one-page **Business Brief** covering items 1 to 6. Your instructor should receive the problem and scope before the discussion.

Phase 2: Solution Design

By the end of this phase, the system exists on paper: five workflows, five owners, and an agreed handoff between them. This phase decides whether integration works later.

Course link: Module 1 (defining input and output), Module 5 (guardrails), Module 7, Session 1 (risk tiers and model routing), Module 8 (evaluation and monitoring).

What to do

1. **Features list.** List what the system does for its user. Mark each feature **Must have** or **Nice to have**. Must-haves are only what the end-to-end demo needs; keep them to eight or fewer.
2. **Split the system into five workflows.** Follow the path of the patient or the document through the system. Each workflow has one clear job, one owner and a similar amount of work.
3. **Workflow Spec Card.** Each owner fills one card for their workflow using Template A2: purpose, trigger, input, steps, AI component, tools, output, human approval point and failure behaviour.
4. **Integration map.** List every handoff between workflows using Template A3: from, to, what is handed over, and the exact fields. The output on one card must match the input on the next, field for field.
5. **Safety plan.** Half a page that answers four questions:
    - Where does your synthetic data come from?
    - Where is the PHI gate, in case real identifiers are typed in?

    - What is the risk tier of each workflow (low, medium, high, critical), and where is the human approval step for high-tier ones?
    - What does the system refuse, and who does it escalate to?
6. **Evaluation and monitoring plan.** Half a page that answers four questions:
    - What does the LLM-as-judge score, and against which rubric?
    - What does each workflow log, and how is it traced in LangSmith?
    - Which numbers appear on the monitoring dashboard?
    - Which alerts fire, at what threshold, and who receives them?

Building blocks from the course

Most workflows combine one or two of these.

Building block	Typical use in a project	Module
Structured prompts and extraction	Turn free text into SOAP, SBAR or JSON records	2
RAG over approved documents	Answer questions with citations from guidelines or leaflets	3
Tools, agents and MCP	Book, look up, notify or update another system	4
Guardrails	PHI gate, injection defence, refusal and escalation	5
Clinical notes and patient communication	Draft notes, plain-language and multilingual instructions	6
Evaluation and monitoring	LLM-as-judge scoring, LangSmith tracing, a monitoring dashboard and alerts	8

Deliverable and checkpoint 2

A **Solution Design** document: features list, five Workflow Spec Cards, integration map, safety plan, and evaluation and monitoring plan.

Phase 3: Build

By the end of this phase, each member has built and tested their own workflow in n8n against its Workflow Spec Card.

Course link: Modules 2, 3, 4 and 6 for the building blocks; Module 3, Session 3 for orchestration, human-in-the-loop and failing safely; Module 8, Session 2 for logging and tracing.

What to do

Steps 1 to 6 are for each workflow owner. Steps 7 and 8 are for the whole team.

1. **Build the simplest version first.** Produce the output agreed on the card, then add nice-to-have features if time allows.
2. **Test with a sample input.** Do not wait for the workflow before yours. Use a sample input in the format agreed on the integration map.
3. **Make it fail safely.** Add error handling, a safe fallback such as "I don't have enough information", and the human approval step named on the card.
4. **Turn on tracing.** Send every run of your workflow to the team's LangSmith project, so each input, model call and output can be inspected.
5. **Test it alone.** Run at least five test cases and record expected against actual output:
    - three normal cases
    - one edge case, such as missing or ambiguous information
    - one unsafe case, such as an input containing identifiers, a prompt injection or an out-of-scope request
6. **Document it on one page.** What it does, how to run it, the prompts used, a screenshot of the n8n workflow, the test results and known limits.
7. **Hold a mid-build sync.** Halfway through the build, each owner shows their real output to the owner of the next workflow.
8. **Control changes.** Any change to an input or output goes through the Integration Owner, who updates the integration map the same day.

Deliverable and checkpoint 3

Five working workflows and **five one-page workflow documents**.

Phase 4: Integration and Testing

By the end of this phase, the five workflows run as one system and you have measured it against the KPIs you set in Phase 1.

Course link: Module 8 (evaluation benchmark, LLM-as-judge, tracing, dashboards and alerting thresholds).

What to do

1. **Connect one handoff at a time.** Follow the integration map in order. Test each handoff before you add the next one.
2. **Write three end-to-end scenarios** with synthetic patients. Each one runs through all five workflows:
    - a typical case
    - a complicated case
    - a case that must trigger a refusal or an escalation to a human
3. **Trace every run.** Check in LangSmith that every step of each scenario is traced, from the first input to the final output.

4. **Score with an LLM-as-judge.** A second model scores every output from 1 to 5 on the five dimensions from the course: factual accuracy, groundedness, safety compliance, format adherence and bias. Give it a specific rubric for each dimension.
5. **Calibrate the judge.** Two members who did not build the workflow score the same scenarios by hand. Compare their scores with the judge's and adjust its rubric until they agree. Fix any safety failure before the pitch.
6. **Build the monitoring dashboard.** One page that shows your KPIs, the judge's scores, failed runs, response time and the number of escalations.
7. **Set alerts.** Define at least one soft threshold that notifies a named owner and one hard threshold that pauses the workflow. Write the reason for each.
8. **Keep an issue log.** Record each problem, its cause, the fix and its status. Use Template A5.
9. **Report KPI results.** Fill the "actual result" column of your KPI table. Report the real numbers; a missed target with a clear explanation scores better than an unsupported claim.
10. **List limitations and next steps.** Say what the system cannot do yet and what a real pilot would need.
11. **Record a backup video** of one full run, in case the live demo fails.

Deliverable and checkpoint 4

A **working end-to-end system** with tracing, a **monitoring dashboard** and alerts; a **Test Report** (scenarios, judge and human scores, issue log, KPI results and limitations); and a **backup video** of one full run.

Phase 5: Business Pitch

By the end of this phase, you have presented the system to a decision-maker, not to engineers. The audience should finish wanting to pilot it.

Course link: Module 1, Session 2 (success metrics: outcomes, cost and workforce well-being).

What to do

1. **Build the story.** Follow the pitch structure below and keep to business language.
2. **Share the speaking.** All five members speak, each on their own workflow.
3. **Prepare the live demo.** Choose one scenario from Phase 4 and apply the tips below.
4. **Prepare for questions.** The pitch ends with questions from the audience.

Pitch structure

Part	What to show
1. Problem	The problem, who has it and what it costs today
2. Audience	The users and the buyer
3. Solution	What the system does, with the five workflows on one slide
4. Live demo	One scenario from start to finish

Part	What to show
5. Results and value	KPI results on the monitoring dashboard, expected against actual, and the business value
6. Safety and limits	Data used, human approval points and current limitations
7. Next steps	What you need to run a pilot

Demo and delivery tips

- Rehearse one scenario until it runs the same way every time.
- Show what the user sees. Keep the n8n canvas on screen only briefly.
- Show the human approval step; it is what makes the system trustworthy.
- Speak in business terms: "saves 15 minutes per discharge", not "uses a RAG pipeline".
- Keep the backup video open in another tab.

Deliverable and checkpoint 5

The **presentation slides** and the **live demo**. The pitch itself is checkpoint 5.

Final deliverables checklist

Submit everything in one shared folder (github/drive) before the pitch.

- ☐ Business Brief (one page)
- ☐ Solution Design: features list, five Workflow Spec Cards, integration map, safety plan, evaluation and monitoring plan
- ☐ Five one-page workflow documents
- ☐ The integrated system: exported n8n workflows, the synthetic test data, the monitoring dashboard and the alert rules
- ☐ Test Report: scenarios, LLM-as-judge and human scores, LangSmith traces, issue log, KPI results, limitations
- ☐ Presentation slides
- ☐ Backup video of one full run

Evaluation rubric

The team earns 60 points together and each member earns 40 points individually, for a total of 100.

Criterion	Scored for	Points	What excellent looks like
Business discovery	Team	15	A specific problem with numbers, measurable objectives, a clear user

Criterion	Scored for	Points	What excellent looks like
			and buyer, KPIs with baselines
Solution design and integration	Team	15	Five workflows of similar weight that run as one system with no manual copying between them
Evaluation, monitoring and safety	Team	15	Three scenarios scored by a calibrated LLM-as-judge, every run traced in LangSmith, a dashboard with working alerts, KPIs reported honestly, human approval and refusal rules working
Pitch and live demo	Team	15	A business story a decision-maker follows, a demo that runs, all five members speaking
Own workflow	Individual	20	Works reliably, matches its Workflow Spec Card and fails safely
Own documentation and tests	Individual	10	One clear page and five recorded test cases, including the unsafe case
Contribution	Individual	10	Meets checkpoints, supports integration and presents their part clearly

Appendix A: Templates

Copy these five templates into your own documents and fill them in.

A1. Problem statement

> ***[Who]*** *in* ***[setting]*** *struggle with* ***[problem]*** *when* ***[situation]****. Today this causes* ***[measurable impact]****. The current workaround is* ***[what people do now]****, which falls short because* ***[reason]****. Solving it would give* ***[benefit to users and to the organisation]****.*

A2. Workflow Spec Card

One card per workflow, filled by its owner.

Field	What to write
Workflow name and owner	A short name and the member responsible
Purpose	One sentence: what this workflow does for the user
Trigger	What starts it: a form, a chat message, a schedule, or the output of another workflow
Input	The fields it receives, their format, and which workflow or user sends them
Steps	Three to seven steps from input to output
AI component	Structured prompt, RAG or agent; which model and why
Tools and data sources	Documents, APIs or systems it reads from or writes to
Output	The fields it produces, their format, and which workflow or user receives them
Risk tier	Low, medium, high or critical
Human approval point	Who reviews what, and before which action
If it fails	The fallback message and who is notified
Logging and tracing	What is logged for each run and where it is traced
Test cases	Three normal, one edge, one unsafe

A3. Integration map

One row per handoff.

From	To	What is handed over	Fields and format	Trigger
Workflow 1	Workflow 2	*The item passed on*	*Exact field names, for example JSON*	*What starts the handoff*

A4. KPI table

KPI	Lens	Baseline today	Target	How it is measured	Actual result
What you measure	*Outcomes, cost or workforce well-being*	*Value before the system*	*Value you aim for*	*Method used in testing*	*Filled in Phase 4*

A5. Issue log

No.	Issue	Workflow	Cause	Fix	Status
1	*What went wrong*	*Where it appeared*	*Why it happened*	*What you changed*	*Open or closed*

Appendix B: Worked example

This example shows the level of detail expected in Phases 1 and 2. The hospital and all numbers are invented; do not reuse it as your project.

Post-Discharge Care Assistant

Business problem. On a busy ward, discharge instructions are generic and nobody follows up with the patient. The problem sits in the discharge and follow-up stage of the patient journey.

Problem statement. Ward nurses and residents in a 200-bed general hospital struggle to prepare clear discharge instructions and follow up with patients when the ward is busy. Today this causes about 20 minutes of paperwork per discharge, and patients go home with questions nobody answers. The current workaround is a generic printed leaflet and no follow-up contact, which falls short because the leaflet is not specific to the patient and nobody checks on them. Solving it would give patients instructions they understand, nurses more time, and the hospital earlier warning of complications.

Business objectives.

- Cut the time to prepare discharge documents from 20 minutes to 5.
- Contact every discharged patient within 48 hours.
- Pass every red-flag symptom to a nurse within one hour.

Target audience. The users are ward nurses and residents. The beneficiaries are discharged patients and their families. The buyer is the hospital's director of nursing.

Success metrics (KPIs).

KPI	Lens	Baseline today	Target	How it is measured
Minutes to prepare discharge documents	Cost	20	5	Timed on 10 synthetic cases
Instructions written at grade 6 to 7 reading level	Outcomes	Not measured	90% of documents	Readability score on each output
Red-flag messages passed to a nurse	Outcomes	0%	100%	10 synthetic patient messages containing red flags
Nurses who say the draft saves them effort	Workforce well-being	Not measured	4 of 5 nurses	Short survey after testing

Scope. In scope: adult patients discharged from one medical ward, with documents in Arabic and English. Out of scope: diagnosis, medication changes and any connection to the hospital's live record system.

Features list. Must have: draft discharge summary, bilingual patient instructions, cited answers to patient questions, follow-up booking with reminders, and red-flag alerts to a nurse. Nice to have: voice messages and a view for family members.

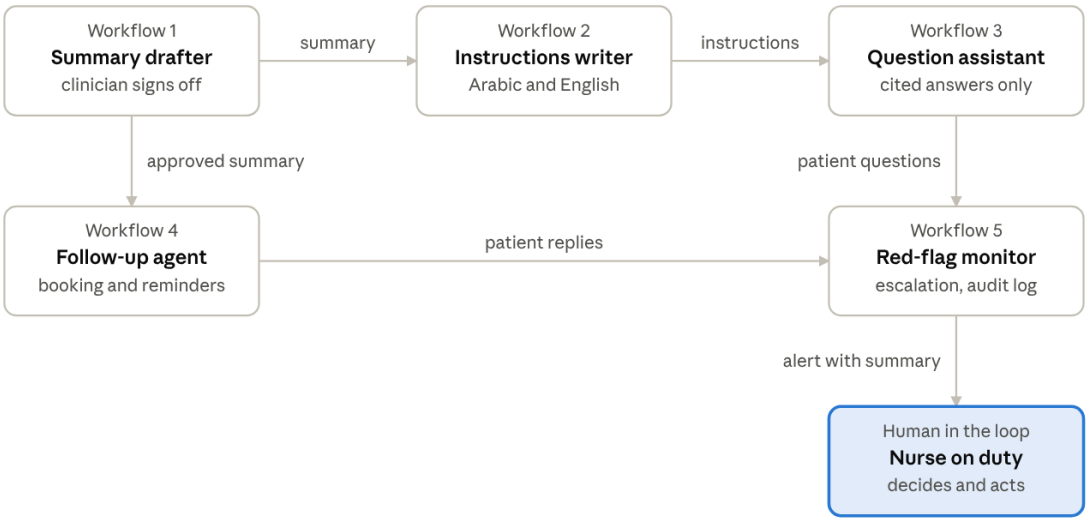
The five workflows.

No.	Workflow	What it does	Input	Output	Course module
1	Summary drafter	Drafts a structured discharge summary for the clinician to edit and sign	Synthetic encounter notes	Approved summary (JSON)	2 and 6
2	Instructions writer	Rewrites the approved summary as plain-language instructions in Arabic and English	Approved summary	Patient instructions	6

No.	Workflow	What it does	Input	Output	Course module
3	Question assistant	Answers patient questions from the instructions and approved hospital leaflets, with citations	Patient question, patient instructions	Cited answer or a refusal	3
4	Follow-up agent	Books the follow-up visit, sends reminders and a 48-hour check-in message	Approved summary	Appointment, reminders, patient replies	4
5	Red-flag monitor	Screens patient messages for red flags, alerts a nurse, logs every action and shows the KPIs	Patient questions and replies	Nurse alert, audit log, dashboard	5 and 8

Integration map.

Each workflow hands its output to the next, and a nurse makes the final call



worked example · 5 workflows, 6 handoffs

The clinician-approved summary from workflow 1 feeds workflows 2 and 4. Everything a patient writes, to the question assistant or to the follow-up agent, passes through the red-flag monitor before a nurse acts.

Safety plan.

- **Data:** 20 synthetic patients written by the team.
- **PHI gate:** at the entry of workflow 1, before any text reaches a model.
- **Risk tiers:** workflows 1 and 5 are high tier, workflows 2 and 3 are medium, workflow 4 is low.
- **Human approval:** a clinician signs the summary, and a nurse approves the instructions before they are sent.
- **Refusals:** the question assistant refuses questions about changing a dose or making a diagnosis, and passes them to the nurse.

Evaluation and monitoring plan.

- **LLM-as-judge:** scores every summary, instruction sheet and answer on the five dimensions, after calibration on 10 cases scored by two team members.
- **Tracing:** every run of all five workflows is traced in LangSmith.
- **Dashboard:** minutes per discharge, reading level, judge scores, escalations and failed runs.
- **Alerts:** a soft alert to the Evaluation Owner when the average safety score falls below 4, and a hard stop that pauses the question assistant when any answer scores 1 on safety.